

computers have a fixed and relatively small amount of memory (RAM) that is used by running applications to store data. Every byte of this memory has an individual address.

When you launch an application on your OS, it gets its own address space that gives it access to a large chunk of memory (4GB on 32bit OSs). These memory addresses however are virtual and don't correspond to the actual address in the RAM. When the application requests access to read or write to a location in one of its virtual addresses, the operating system translates that virtual address to its actual location and performs the read / write operation there.

This location that actual data is stored is managed by the virtual memory system of the OS, and the real location could be the RAM or even the swap partition / page file. Here the operating system is virtualising the memory, while the application runs as if it is working directly with the memory, it is actually the operating system that is intercepting every read and write and directing it to the correct location.

There are plenty of advantages to this. For one, it allows the OS to use your hard drive space for storing data when your RAM runs out of space. Secondly it allows each application process to have isolated access to the memory, thus improving security. Applications also need to be written to manage a memory space shared with other applications as they get isolated instances of the same memory, and leave the hard work to the OS.

Another less apparent example of virtualisation in action is Intel's Hyper-threading technology. If you haven't heard of this before, here is a gist: It is a feature present in some Intel processors that doubles the number of processors that appear to be present in the computer. So for a quad-core machine, your OS will show eight processors, for an eight-core machine, it will show sixteen, and so on.

Obviously there is no black magic behind this. The processor simply includes technology that gives the appearance there being double the number of processors then there really are. Intel's reasoning for doing so it to improve the performance of multiple applications running in parallel by giving each processor double the work in its pipeline.

We'll leave out the details of how it works, since it's out of the current scope. Instead let's reiterate that this isn't magic, and Intel hasn't doubled the processor core count out of the goodness of their heart. This is a virtualisation technology working at a very low level that shows the computer two virtual processors for each real processor.